Additional Documentation

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1 Introduction

This document will serve as additional documentation for my Capstone project "Can an Unsupervised Temporal Analysis Provide Useful Magic: The Gathering Data"

Its purpose is solely to provide more examples of tracking communities over time. As referenced in the "Tracking Communities Overtime" subsection within the "Discussion" section

2 Universal Tables

This section will display tables that may be referenced in the chapters below.

Community	SS1	SS2	SS3	SS4	SS5	SS6	SS7	SS8
0	59	89	30	41	21	44	85	90
1	54	36	34	21	25	63	45	18
2	49	20	67	105	56	32	25	85
3	60	43	33	59	54	98	107	80
4	11	91	23	11	16	1	32	92
5	11	36	26	10	39	49	73	45
6	10	100		14		29	5	
7	40			57		33	88	
8	71			71			64	
9	38			34				

Table 1: Number of cards in each snapshots communities

Table 1 shows for each community in the Snapshot how many unique cards were present within it, (empty space meaning that snapshot did not have that many communities). Community 4 in snapshot 6 had only a single card which is not substantial enough information to represent any decks within the metagame so that community was dropped. For the remainder of this analysis snapshot 6 will be treated as having 7 total communities. While the next lowest card count is community 6 in snapshot 7 with 5, however a deck can contain 4 copies of the same card and as this table only represents unique cards. With 20 cards, this community can represent a strategy, with the remaining slots in the deck being dedicated to lands and meta staples. For this reason this reason and for it being only a singular occurrence within the entire analysis community 4 in snapshot 6 was dropped.

Snapshot	Number of cards lost
1	150
2	239
3	49
4	285
5	27
6	79
7	213

Table 2: Number of cards that fell out of the meta

Table 2 shows the number of cards that did not show up in subsequent snapshots (cards that did not achieve any tournament wins and therefore fell out of the meta)(e.g 285 cards where lost in the jump from snapshot 4 to 5).

Community	SS2	SS3	SS4	SS5	SS6	SS7	SS8
0	34	11	14	6	24	25	23
1	14	4	21	1	40	32	2
2	14	7	53	21	9	7	14
3	16	7	32	17	42	50	28
4	43	6	10	9		14	20
5	10	2	9	19	11	30	12
6	31		14		14	5	
7			50		24	52	
8			40			39	
9			16				
Total	162	37	259	73	165	254	99

Table 3: Communities new cards joined

Table 3 showcases the number of cards that were not present in the previous snapshot, that saw competitive play in the current snapshot and to which community they joined. Note it is possible for a card to join the metagame, leave and rejoin at a later point. This table simply counts cards not seen in the previous snapshot.

Table 4 showcases the Jaccard similarity between Snapshots 1 and 2.

Table 5 showcases the Jaccard similarity between Snapshots 2 and 3.

Table 6 showcases the Jaccard similarity between Snapshots 3 and 4.

Table 7 showcases the Jaccard similarity between Snapshots 4 and 5.

Table 8 showcases the Jaccard similarity between Snapshots 5 and 6.

Table 9 showcases the Jaccard similarity between Snapshots 6 and 7.

	Number of cards in community	Com 0	Com 1	Com 2	Com 3	Com 4	Com 5	Com 6	Cards accounted for	Cards unaccounted for
Com 0	59	0.021	0.145	0	0.291	0.007	0.022	0.006	42	17
Com 1	54	0.265	0.047	0	0	0.007	0.011	0.013	38	16
Com 2	49	0.104	0	0.062	0	0	0.09	0	24	25
Com 3	60	0	0	0	0.01	0.28	0	0.019	37	23
Com 4	11	0.02	0.022	0	0	0.085	0	0	11	0
Com 5	11	0	0	0	0.038	0	0	0	2	9
Com 6	10	0	0.022	0.071	0	0	0	0	3	7
Com 7	40	0.157	0.056	0	0	0.008	0	0.157	26	14
Com 8	71	0.032	0	0	0	0.025	0.009	0.336	53	18
Com 9	38	0	0	0	0.013	0	0.254	0.007	17	21

Table 4: Jaccard similarity table between snapshots 1 and 2

	Number of cards in community	Com 0	Com 1	Com 2	Com 3	Com 4	Com 5	Cards accounted for	Cards unaccounted for
Com 0	89	0.092	0	0	0.017	0	0.055	18	71
Com 1	36	0.015	0	0	0.078	0	0.35	22	14
Com 2	20	0	0	0	0	0	0	0	20
Com 3	43	0	0.013	0.019	0.333	0	0.015	23	20
Com 4	91	0.034	0.008	0.4	0	0	0	50	41
Com 5	36	0.066	0.0145	0	0	0.283	0.016	19	17
Com 6	100	0	0.252	0.084	0	0.034	0	44	56

Table 5: Jaccard similarity table between snapshots 2 and 3

	Number of cards in community	Com 0	Com 1	Com 2	Com 3	Com 4	Com 5	Com 6	Com 7	Com 8	Com 9	Cards accounted for	Cards unaccounted for
Com 0	30	0.145	0	0.098	0	0	0	0	0.024	0	0.016	24	6
Com 1	34	0.027	0	0.007	0	0	0	0	0	0.364	0.015	32	2
Com 2	67	0	0	0.293	0.016	0	0.013	0	0.033	0.022	0	49	18
Com 3	33	0.28	0	0	0.022	0.023	0	0	0	0	0.015	20	13
Com 4	23	0	0	0	0.012	0	0	0	0.013	0	0.357	17	6
Com 5	26	0	0	0	0.349	0	0	0	0	0	0	22	4

Table 6: Jaccard similarity table between snapshots 3 and 4

	Number of cards in community	Com 0	Com 1	Com 2	Com 3	Com 4	Com 5	Cards accounted for	Cards unaccounted for
Com 0	41	0	0	0.032	0	0	0.212	17	24
Com 1	21	0	0	0.013	0	0	0	1	20
Com 2	105	0.008	0	0.073	0.252	0.017	0	46	59
Com 3	59	0	0.377049	0	0.018	0	0.01	26	33
Com 4	11	0	0.029	0	0	0	0.042	3	8
Com 5	10	0	0	0.015	0.016	0	0.021	3	7
Com 6	14	0	0	0	0	0	0	0	14
Com 7	57	0	0	0.037	0	0.074	0.021	11	46
Com 8	71	0.179	0	0.024	0.016	0	0	19	52
Com 9	34	0	0	0.154	0	0	0	12	22

Table 7: Jaccard similarity table between snapshots 4 and 5

	Number of cards in community	Com 0	Com 1	Com 2	Com 3	Com 5	Com 6	Com 7	Cards accounted for	Cards unaccounted for
Com 0	21	0.413	0	0	0	0	0	0	19	2
Com 1	25	0	0	0.583	0.008	0.014	0	0	23	2
Com 2	56	0	0.214	0	0.069	0	0.214	0.023	48	8
Com 3	54	0.01	0.017	0.024	0.394	0	0	0.012	49	5
Com 4	16	0	0	0	0.018	0.083	0	0.14	13	3
Com 5	39	0	0	0	0	0.571	0	0	32	7

Table 8: Jaccard similarity table between snapshots 5 and 6

	Number of cards in community	Com 0	Com 1	Com 2	Com 3	Com 4	Com 5	Com 6	Com 7	Com 8	Cards accounted for	Cards unaccounted for
Com 0	44	0.016	0	0	0.013	0	0.315	0	0.008	0	33	11
Com 1	63	0.007	0.009	0	0.172	0.203	0	0	0.007	0.008	45	18
Com 2	32	0.009	0	0.018	0.015	0.016	0.019	0	0.008	0.231	26	6
Com 3	98	0.326	0.007	0.15	0.01	0.008	0.062	0	0.005	0.025	80	18
Com 5	49	0	0	0.0137	0.006	0	0.008	0	0.305	0.009	36	13
Com 6	29	0	0	0	0.183	0	0.02	0	0	0.011	24	5
Com 7	33	0.103	0.164	0	0.022	0	0	0	0	0	25	8

Table 9: Jaccard similarity table between snapshots 6 and 7

	Number of cards in community	Com 0	Com 1	Com 2	Com 3	Com 4	Com 5	Cards accounted for	Cards unaccounted for
Com 0	85	0.012	0.01	0.223	0.0123	0.06	0.008	47	38
Com 1	45	0.007	0	0.12	0.008	0.015	0	18	27
Com 2	25	0.009	0	0.089	0	0	0	10	15
Com 3	107	0.359	0	0.005	0.016	0.021	0.02	63	44
Com 4	32	0.043	0	0.009	0	0.117	0.013	20	12
Com 5	73	0.019	0	0.046	0	0.32	0.009	51	22
Com 6	5	0	0	0.011	0	0	0	1	4
Com 7	88	0.006	0	0.036	0.333	0.006	0.023	53	35
Com 8	64	0.013	0.224	0.007	0.029	0.013	0.282	48	16

Table 10: Jaccard similarity table between snapshots 7 and 8

Table 10 showcases the Jaccard similarity between Snapshots 7 and 8.

3 Tracking Snapshot 1 Community 0

In the main paper it was shown Snapshot 1's Community 0 surviving and becoming Snapshot 2's Community 3. Going onwards from snapshot it can be continued to be tracked.

Using Table 5 we can keep track of the new community 3. Here it is shown that 0.333 precent of card continue into snapshot 3 and end up in community 3. This is backed up by the tacking the most important cards of community 3.

SS2 Community 3	SS3 Community 3
"Screaming Nemesis" 0.006	"Monstrous Rage" 0.027
"Heartfire Hero" 0.006	"Burst Lightning" 0.016
"Emberheart Challenger" 0.006	"Screaming Nemesis" 0.015
"Manifold Mouse" 0.006	"Emberheart Challenger" 0.013
"Hired Claw" 0.005	"Heartfire Hero" 0.013
"Witchstalker Frenzy" 0.005	"Magebane Lizard" 0.013
"Lightning Strike" 0.003	"Manifold Mouse" 0.013
"Magebane Lizard" 0.003	"Hired Claw" 0.012
"Trumpeting Carnosaur"0.001	"Monastery Swiftspear" 0.009
"Sunspine Lynx"0.001	"Witchstalker Frenzy" 0.009

Table 11: Top Cards by PageRank

Seen again is the same communities individuals surviving and remaining important within their new community. Here this community can be determined to have survived another snapshot. Table 12 showcases this.

From here a it can be determined that snapshot 3 Community 3 becomes community 0 in snapshot 4. While some of the highly ranked cards from this community were actually banned, ("Monstrous Rage", "Heartfire Hero", "Magebane Lizard"), the remaining cards stuck together and remained important.

SS3 Community 3	
"Monstrous Rage" 0.027	"Screaming Nemesis" 0.005
"Burst Lightning" 0.016	"Overprotect" 0.005
"Screaming Nemesis" 0.015	"Pawpatch Recruit" 0.004
"Emberheart Challenger" 0.013	"Twinmaw Stormbrood" 0.004
"Heartfire Hero" 0.013	"Burst Lightning" 0.004
"Magebane Lizard" 0.013	"Hired Claw" 0.004
"Manifold Mouse" 0.013	"Emberheart Challenger"0.003
"Hired Claw" 0.012	"Innkeeper's Talent"0.003
"Monastery Swiftspear" 0.009	"Monastery Swiftspear" 0.002
"Witchstalker Frenzy" 0.009	"Witchstalker Frenzy" 0.002

Table 12: Top Cards by PageRank

This should be a clear indication as how to track a community over time. Step by step it is analysed verifying where the community ended up within each snapshot.

4 Example of a community dying

In the jump from Snapshot 4 to snapshot 5, shown on Table 7 community 6 in Snapshot 4 had 14 cards 0 of which survived into the next snapshot meaning this community died out.